

JUNIT BASIC TESTING EXERCISE

EXERCISE 1: SETTING UP JUNIT

Calculator.java

```
public class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
    public int subtract(int a, int b) {  
        return a - b;  
    }  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
}
```

CalculatorTest.java

```
import org.junit.Test;  
import static org.junit.Assert.*;  
public class CalculatorTest {  
    @Test  
    public void testAdd() {  
        Calculator calc = new Calculator();  
        assertEquals(8, calc.add(5, 3));  
    }  
  
    @Test  
    public void testSubtract() {  
        Calculator calc = new Calculator();  
        assertEquals(2, calc.subtract(5, 3));  
    }  
}
```

@Test

```
public void testMultiply() {
```

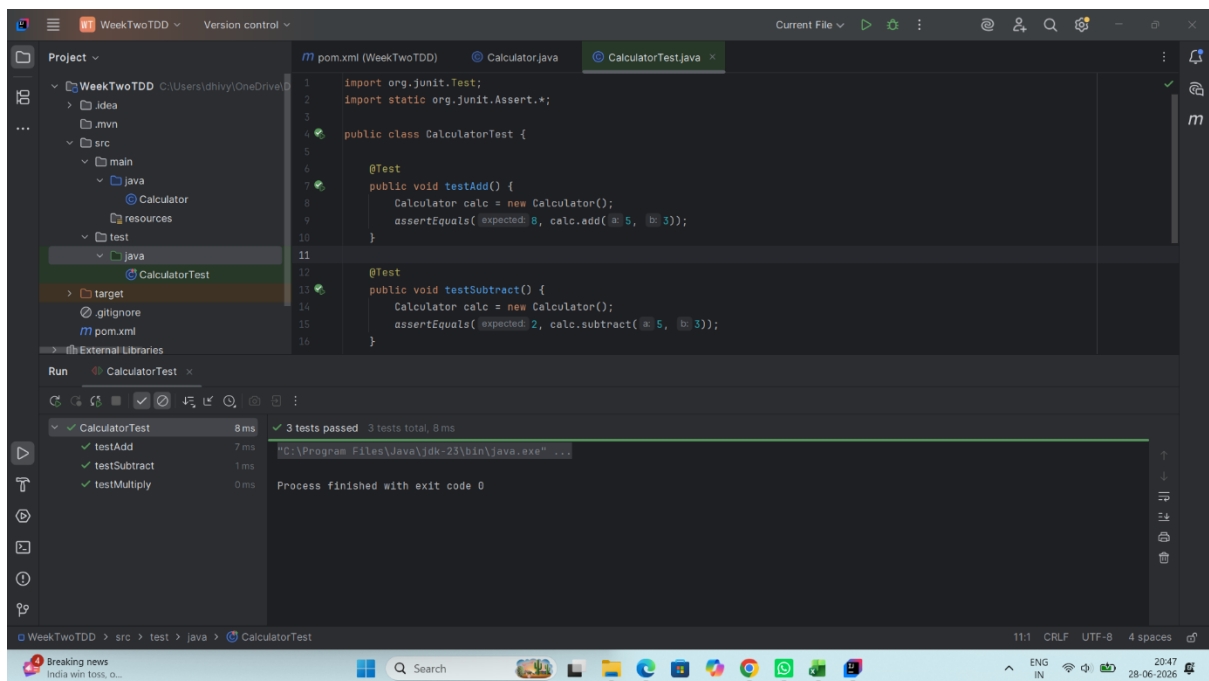
```
    Calculator calc = new Calculator();
```

```
    assertEquals(15, calc.multiply(5, 3));
```

```
}
```

```
}
```

OUTPUT

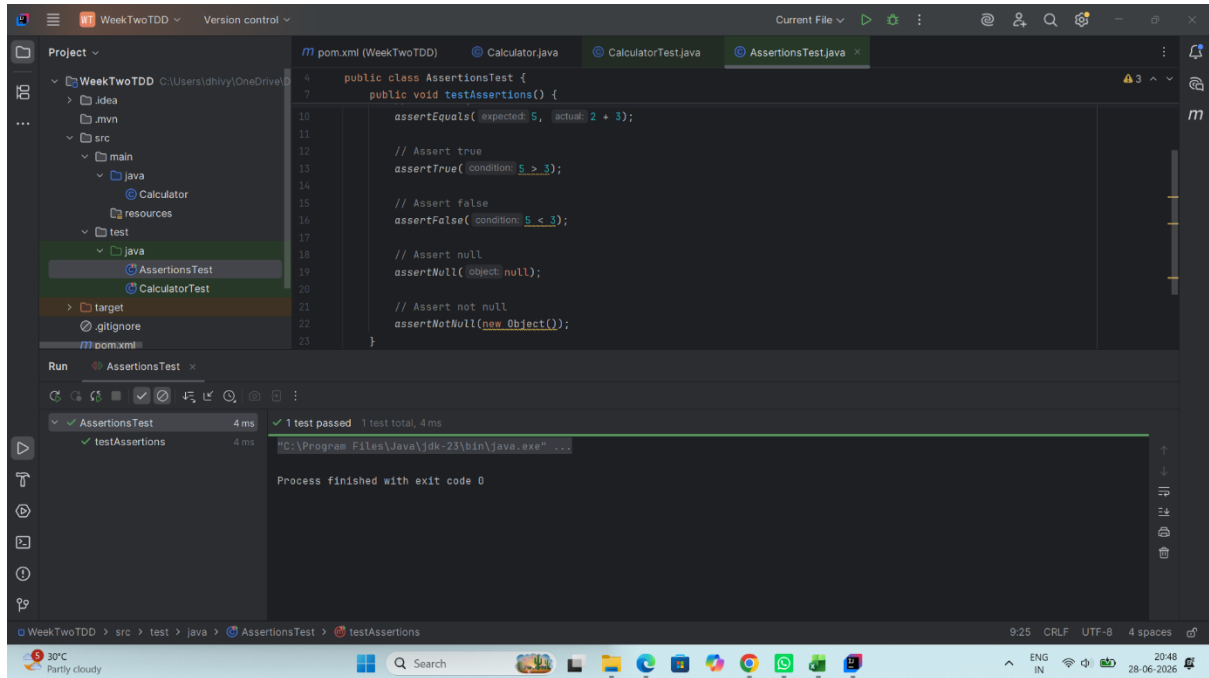


EXERCISE 3: ASSERTIONS IN JUNIT

AssertionsTest.java

```
import org.junit.Test;
import static org.junit.Assert.*;
public class AssertionsTest {
    @Test
    public void testAssertions() {
        // Assert equals
        assertEquals(5, 2 + 3);
        // Assert true
        assertTrue(5 > 3);
        // Assert false
        assertFalse(5 < 3);
        // Assert null
        assertNull(null);
        // Assert not null
        assertNotNull(new Object());
    }
}
```

OUTPUT



EXERCISE 4: ARRANGE-ACT-ASSERT (AAA) PATTERN, TEST FIXTURES, SETUP AND TEARDOWN METHODS IN JUNIT

AAAPatternTest.java

```
import org.junit.Before;
import org.junit.After;
import org.junit.Test;
import static org.junit.Assert.*;
public class AAAPatternTest {
    Calculator calc;
    @Before
    public void setUp() {
        calc = new Calculator();
        System.out.println("Setup done - Calculator created");
    }
    @After
    public void tearDown() {
        calc = null;
        System.out.println("Teardown done - Calculator destroyed");
    }
    @Test
    public void testAddAAA() {
        // Arrange
        int a = 10;
        int b = 5;
        // Act
        int result = calc.add(a, b);
        // Assert
        assertEquals(15, result);
    }
}
```

```
}  
@Test  
public void testSubtractAAA() {  
    // Arrange  
    int a = 10;  
    int b = 5;  
    // Act  
    int result = calc.subtract(a, b);  
    // Assert  
    assertEquals(5, result);  
}  
@Test  
public void testMultiplyAAA() {  
    // Arrange  
    int a = 10;  
    int b = 5;  
    // Act  
    int result = calc.multiply(a, b);  
    // Assert  
    assertEquals(50, result);  
}  
}
```

OUTPUT

```
1 import org.junit.Before;
2 import org.junit.After;
3 import org.junit.Test;
4 import static org.junit.Assert.*;
5
6 public class AAAPatternTest {
7
8     Calculator calc;
9
10    @Before
11    public void setUp() {
12        calc = new Calculator();
13        System.out.println("Setup done - Calculator created");
14    }
15
16    @After
```

Run AAAPatternTest

Test	Duration	Status
testMultiplyAAA	9 ms	Passed
testAddAAA	1 ms	Passed
testSubtractAAA	1 ms	Passed

3 tests passed 3 tests total, 11 ms

Process finished with exit code 0